

ILIES BIANCA IZABELA

3rd-Year Computer Science & Information Technology Student Focused on Embedded & Digital Hardware Systems

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EDUCATION

Technical University of Cluj-Napoca

Computer Science and Information Technology

2023 – Ongoing

- Relevant coursework: Systems Theory, Electronic Measurements & Sensors, Microprocessor Design, Digital Systems Design, Analog & Digital Circuits, Operating Systems
- Strong background in Microprocessor Design, Digital Systems & Analog Circuits
- Merit-based scholarship recipient (Sem. 1–2, Year 2; Sem. 1, Year 3)

PERSONAL PROJECTS

Zynq SoC Hardware–Software Accelerator

VHDL, Vivado, Vitis, AXI, Q16.16 Fixed-Point

- Designed custom VHDL IP for fixed-point computation (Q16.16) of $a \cdot x + b \cdot y + c$
- Integrated PS–PL communication via AXI GPIO
- Developed C control application in Vitis with UART interface
- Validated via simulation and full hardware–software integration on Zynq-7020

Smart Parking Embedded System

Embedded C/C++, SPI, I2C, UART, PWM, RFID, Ultrasonic

- Integrated RFID (SPI), Ultrasonic sensing, Bluetooth control and LCD display on Arduino
- Developed embedded control logic for secure access and automatic barrier actuation (PWM)
- Managed concurrent SPI, I2C and SoftwareSerial communication
- Implemented real-time parking slot management with event debouncing logic

FPGA Serial Communication Protocol

VHDL, FSM Design, Shift Registers, XOR Checksum, Testbench, Nexys FPGA

- Implemented custom VHDL protocol with start detection and XOR checksum validation
- Designed Generator/Detector FSM architecture with shift registers
- Verified functionality via Vivado simulation and hardware deployment on Nexys FPGA

Travel Assistant App – PoliHack (48h Hackathon)

React, Firebase, JavaScript, Team of 4

- Built full-stack travel platform in 4-member team under 48h deadline
- Implemented authentication, map-based route exploration and real-time chat
- Integrated Firebase backend with dynamic user roles and profile management

INTERNATIONAL EXPERIENCE

Erasmus+ Mobility

App Design Training (60h)

06.12 – 17.12

Paphos, Cyprus

- Practical training in application development
- Multicultural teamwork environment

Erasmus+ VET Mobility

Informatica Reparnet

30.01 – 17.02

Granada, Spain

- Professional IT training in web development and hardware maintenance
- Laptop diagnostics, component servicing and thermal management procedures

TECHNICAL SKILLS

C C++ Python Java

VHDL Zynq AXI SPI I2C

UART PWM

Vivado Vitis Git

React Firebase PostgreSQL

LANGUAGES

Romanian - native

English - B2 (Cambridge)

German - B1

French - A1